EXPERIMENT : 13

Comparative Analysis of NB-IoT and LTE-M for an IoT Sensor Application

OBJECTIVES :

1. To study the operating principles and key characteristics of Narrowband Internet of Things (NB-IoT) and LTE for Machines (LTE-M).

2. To compare NB-IoT and LTE-M in terms of data rate, latency, coverage, energy consumption, and packet delivery ratio.

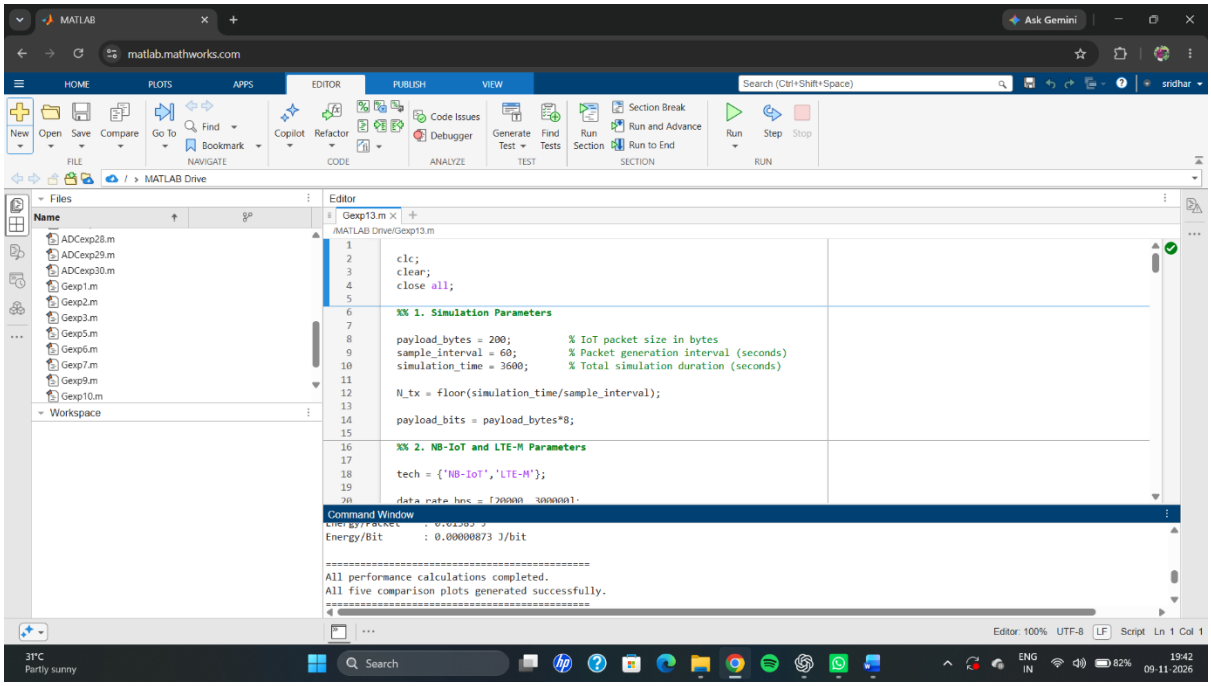
3. To simulate an IoT sensor application that periodically transmits temperature, humidity, or environmental sensor readings.

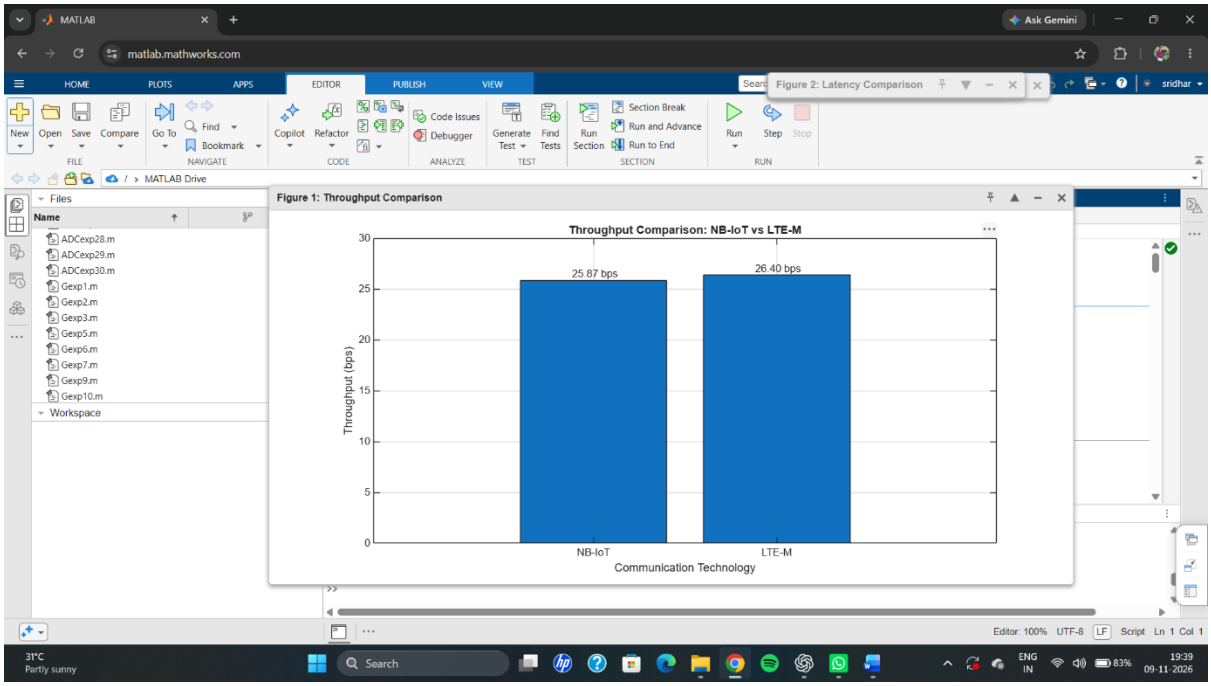
4. To evaluate the impact of packet size, transmission interval, signal quality, and retransmissions on network performance.

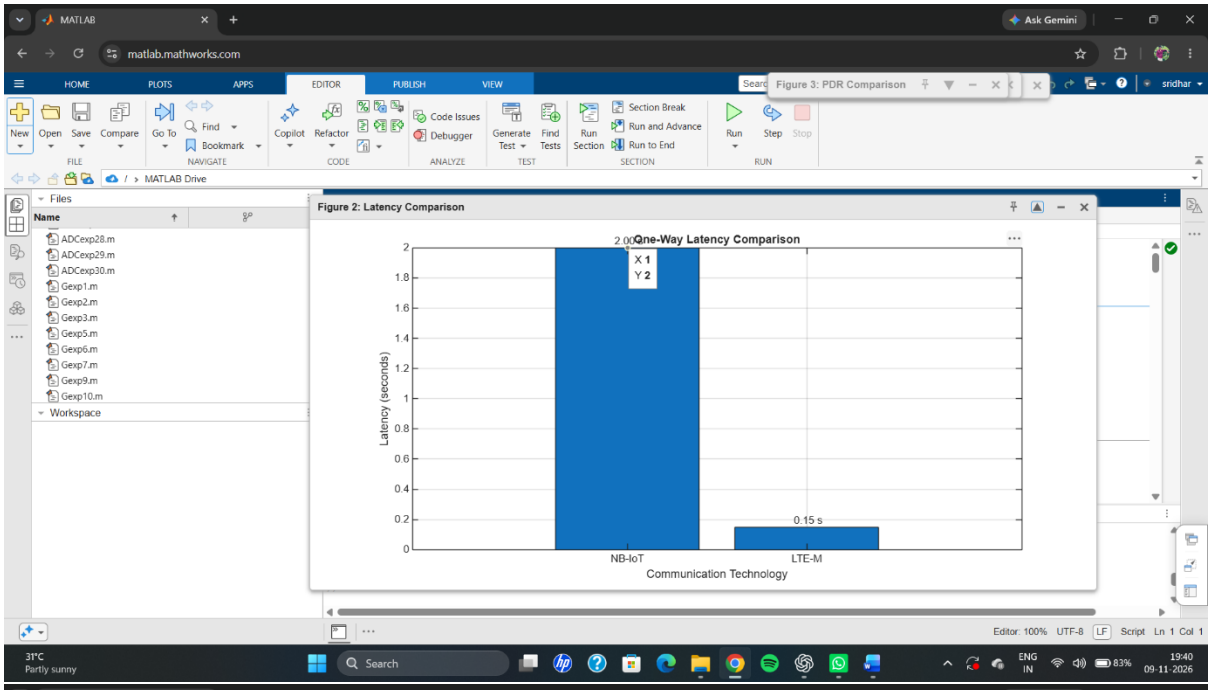
5. To analyze the suitability of NB-IoT and LTE-M for different IoT applications.

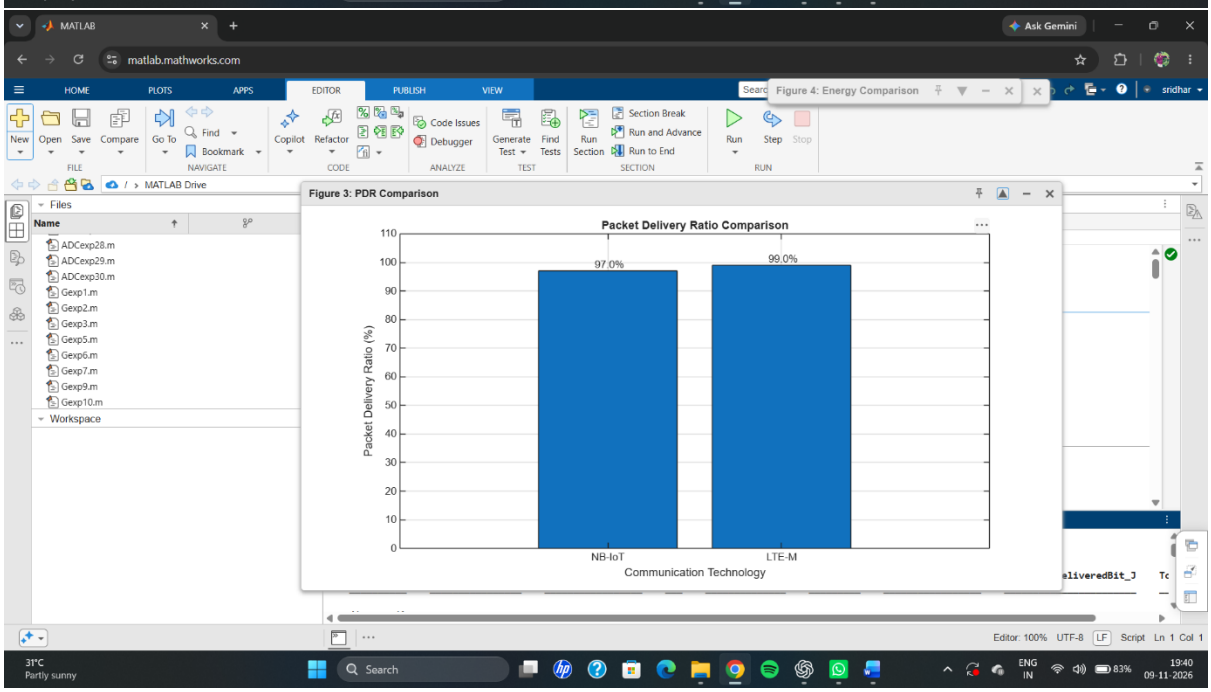
6. To plot and interpret the performance metrics of both technologies using MATLAB.2.

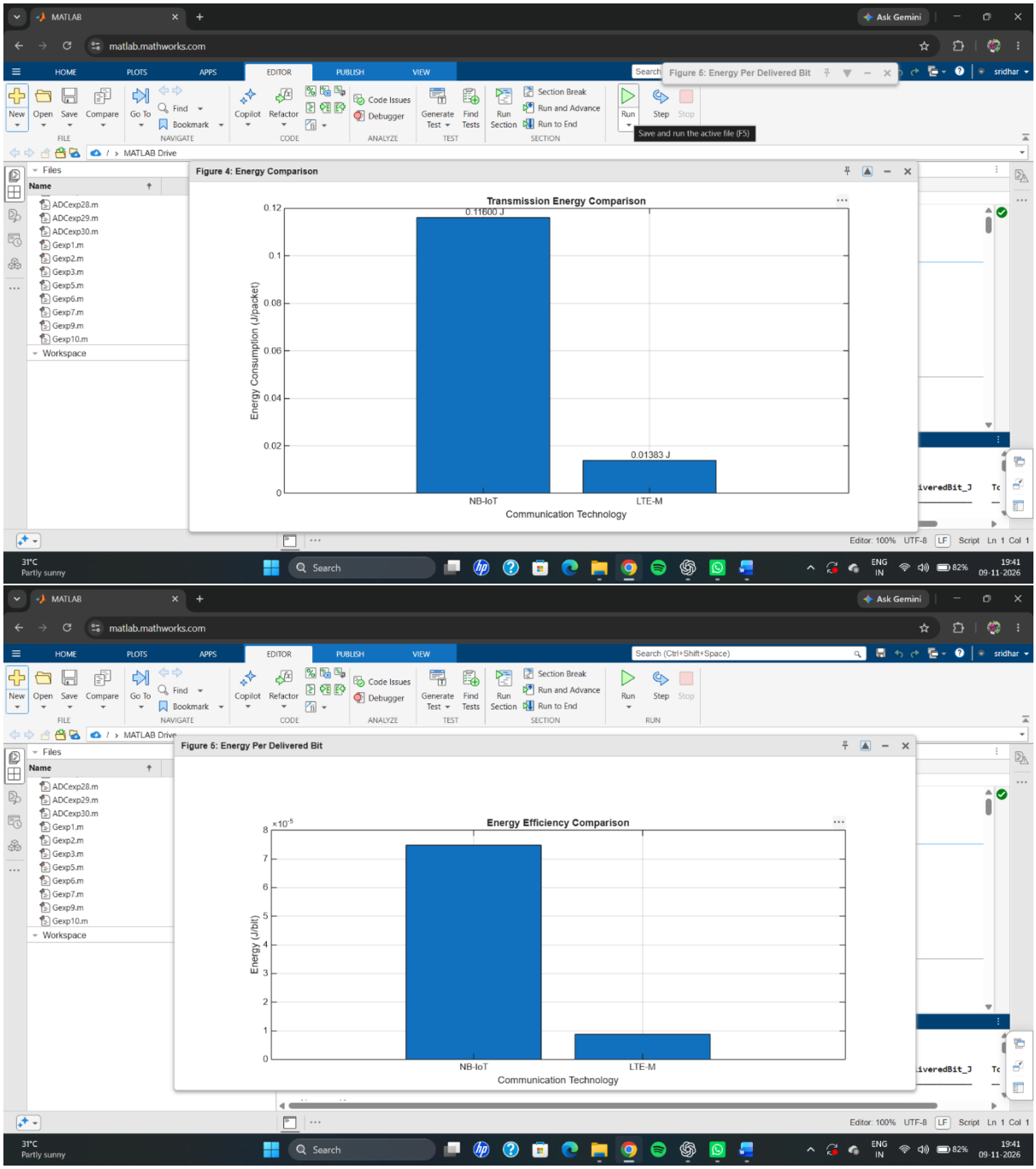
PROGRAME CODE :




MATLAB
matlab.mathworks.com
Ask Gemini
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
Figure 2: Latency Comparison
sridhar
New
Open
Save
Compare
Go To
Find
Bookmark
Copilot
Refactor
Code Issues
Debugger
Generate Test
Find Tests
Run Section
Section Break
Run and Advance
Run to End
Run
Step
Stop
FILE
NAVIGATE
CODE
ANALYZE
TEST
SECTION
RUN
MATLAB Drive
Files
Name
ADCexp28.m
ADCexp29.m
ADCexp30.m
Gexp1.m
Gexp2.m
Gexp3.m
Gexp5.m
Gexp6.m
Gexp7.m
Gexp9.m
Gexp10.m
Workspace
Figure 1: Throughput Comparison
Throughput Comparison: NB-IoT vs LTE-M
25.87 bps
26.40 bps
Throughput (bps)
0
5
10
15
20
25
30
NB-IoT
LTE-M
Communication Technology
Editor: 100%
UTF-8
LF
Script
Ln 1 Col 1
31°C
Partly sunny
Search
ENG
IN
83%
19:39
09-11-2026


Figure 2: Latency Comparison
One-Way Latency Comparison
X 1
Y 2
Latency (seconds)
0.15 s
NB-IoT
LTE-M
Communication Technology


Figure 3: PDR Comparison
Packet Delivery Ratio Comparison
Packet Delivery Ratio (%)
97.0%
99.0%
NB-IoT
LTE-M
Communication Technology



RESULT : The MATLAB simulation comparing NB-IoT and LTE-M technologies was successfully implemented and analyzed based on throughput, latency, Packet Delivery Ratio (PDR), and energy consumption. The results showed that LTE-M provides higher throughput, lower latency, and better packet delivery performance, while also consuming less modeled transmission energy per packet than NB-IoT under the given simulation parameters.